HINTS AND SOLUTIONS

Q.1 B
Q.2 C
Q.3 B

Sol. $\frac{4L}{5} = \lambda \Rightarrow \lambda = 8cm$

thus 2 cm corresponds to $\Delta\phi = z/2$
1 cm corresponds to $\Delta\phi = z/4$

So $y = Asm\ \pi/4 = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$

Q.4 B
Q.5 A
Q.6 A

Sol. At the open end pressure node is present. So that
at $x = 0$ P should be zero and $v = f\lambda$,

for first over tone $\frac{3\lambda}{4} = 0.80$

$\Rightarrow$ $f = 300$
$\Rightarrow$ $\omega = 2\pi f = 600\ t$

Hence $P = A \sin \frac{15\pi}{8} x \cos 600\ t$

Q.7 B

Sol.$_{\text{ANU}}$ Higher pressure $\Rightarrow$ higher density

Q.8 C
Q.9 A

Sol. $\lambda_A = \left(10u - \frac{u}{2}\right)\frac{1}{f} = \frac{9.5u}{f}$

$\lambda_B = \left(10u + \frac{u}{2}\right)\frac{1}{f} = \frac{10.5u}{f}$

$\frac{\lambda_A}{\lambda_B} = \frac{\left(\frac{19}{2}\right)\left(\frac{u}{f}\right)}{\left(\frac{21}{2}\right)\left(\frac{u}{f}\right)} = \frac{19}{21}$ **Ans.**

Q.10 B

Sol. A standing wave is formed by two identical waves travelling in opposite. Thus energy stored between two nodes in a standing wave is

$$E = 2[\text{Energy stored in a distance } \frac{n\lambda}{2} \text{ of a travelling wave}]$$

but $\frac{n\lambda}{2} = \frac{n}{2}\cdot\frac{2\pi}{k} = \frac{n\pi}{k}$

So, $E = 2\,(\text{energy density})\,(\text{volume})$

$$= 2\left(\frac{1}{2}\rho A^2\omega^2\right)\left(\frac{Sn\pi}{k}\right)$$

$$= \frac{n\rho A^2\omega^2\pi S}{k}$$

Comparing with given energy, $n = 3$

Q.11 ABC

Q.12 D

Sol. $\lambda = 1\text{m}$

$$v = 400 \text{ m/s} = \sqrt{\frac{T}{0.05}}$$

$$16 \times 10^4 \times \frac{5}{10.20} = T$$

$T = 8000$ N

$v = A\omega$

$80\pi = A \times 2\pi \times 400$

$A = 10$ cm

Q.13 AC

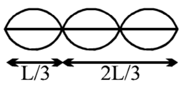
Sol. (A) $w' = \frac{C}{C-v}\omega$ $\qquad \frac{\omega'}{k'} = C \qquad \frac{C}{\omega'}$

(C) $\omega' = \frac{C+v}{C}\,\omega$

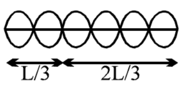
k is same because λ is same.

Q.14 AD

Sol. $\frac{3\lambda}{2} = L$



Minimum loops = 2 $\qquad$ Fundamental $\nu = \frac{3}{2L}V$



Next higher loops = 5

Q.15 ACD

Q.16 AB

Sol. (A) speed $= \frac{\omega}{k} = \frac{10}{1} = 10$ m/s

(B) $2\pi = k = \frac{2\pi}{\lambda}$ $\qquad \lambda = 1$m

$$\phi = \frac{2\pi}{\lambda}\left(\frac{1}{6}\right) = \frac{\pi}{3} = 60°$$

(C) $V_{P\max} = \omega A = 20\pi \times \frac{1}{100} = \frac{\pi}{5}$ m/s

(D) $\phi = 2\omega t$ $\qquad \phi = 20\pi \times \frac{1}{20} = \pi$

Q.17 [Ans. (A)-Q (B)-Q (C)-Q (D)-S]

Sol. (A) and (B) B is displacement node

$\Rightarrow$ $a = 0$, $v = 0$ and $E = 0$ and deformation maximum $\Rightarrow$ PE max.

(D) B is displacement antinods $\Rightarrow$ $a = \max$, KE = 0, PE = 0, V = 0

Q.18 [Ans. (A) P, S; (B) P, R, S; (C) Q, (D) Q, R]

Q.19 [Ans. 0010]

Sol. $20 = 10\log_{10}\left(\frac{I_1}{I_2}\right) \Rightarrow I_1 = 100\, I_2 \quad \frac{p_{01}}{p_{02}} = 10$

Q.20 [Ans. 2]

Sol. $\frac{\lambda}{4} = \ell + e = \ell + 0.6\,r$

$\lambda = 4\ell + 2.4\,r$

$$f = \frac{c}{\lambda} = \frac{c}{4\square + 2.4\,r}$$

$$f_b = \frac{c}{4\square + 2.4\,r_1} - \frac{c}{4\square + 2.4\,r_2}$$

$$= \frac{c\,(2.4\,(r_2 - r_1))}{(4\square + 2.4\,r_1)(4\square + 2.4\,r_2)} = \frac{c \times 2.4 \times 5 \times 10^{-2}}{4.12 \times 4.24} = \frac{12 \times 3.2}{4.12 \times 4.24} \simeq 2\,\text{Hz}$$

Q.21 [Ans. max. = 442 , min. = 255]